SQL QUESTION

SCENARIO BASED

Alternate Sorting

Display Male & Female employees in alternating order.

📊 INPUT TABLE: EMPLOYEE DATA

ID	EmpName	Gender
1	Allen	MALE
2	Benjamin	MALE
3	Charle	MALE
4	Daisy	FEMALE
5	Emma	FEMALE

The Goal: Sort the output to alternate genders:
MALE, FEMALE, MALE, FEMALE, MALE...

Anandnarayanan S | Follow for more SQL Patterns

LOGIC

Partition & Rank

How to create an alternating pattern

1️⃣

Rank within Gender

Use `ROW_NUMBER()` to assign a rank to each employee **within** their gender group.

Allen (M) → Rank 1
Daisy (F) → Rank 1

2️⃣

Two-Level Sort

Sort by Rank first. This pairs the #1s, then the #2s.

Then Sort by Gender DESC. Since 'M' comes after 'F', DESC puts MALE first.

Anandnarayanan S | Follow for more SQL Patterns

SOLUTION

Common Table Expression

alternate_sort.sql

```sql
WITH Ranked_Employees AS (
    SELECT
        id,
        EmpName,
        Gender,
        -- Rank employees per gender group
        ROW_NUMBER() OVER (
            PARTITION BY Gender ORDER BY id
        ) AS gender_rank
    FROM employees
)
SELECT id, EmpName, Gender
FROM Ranked_Employees
-- 1. Sort by Rank (1, 1, 2, 2...)
-- 2. Sort by Gender DESC (Male, Female...)
ORDER BY gender_rank, Gender DESC;
```

Anandnarayanan S | Follow for more SQL Patterns

EXECUTION

Behind the Scenes

How the sorting logic executes

STEP 2: THE FINAL SORT ORDER

Name	Gender	Rank	Sort Order
Allen	MALE	1	1. Rank 1 + M
Daisy	FEMALE	1	2. Rank 1 + F
Benjamin	MALE	2	3. Rank 2 + M
Emma	FEMALE	2	4. Rank 2 + F
Charle	MALE	3	5. Rank 3 + M

💡 **Pro Tip:** DESC on text sorts Z → A.
Since 'M' > 'F', Male appears before Female.

Anandnarayanan S | Follow for more SQL Patterns

WRAP UP

Did you learn something new?

Let your friends learn this too!

Repost to your Friends

Anandnarayanan S | Follow for more SQL Patterns